MTH 101

Calculus & Analytical Geometry

Complete Typed Notes

Part 1B: Lessons 07–12

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1 Lesson 07: Operations on Functions

Definition (Operations on Functions):

Operation	Definition
Sum	$(f+g)(x) = f(x) + g(x)$
Difference	$(f-g)(x) = f(x) - g(x)$
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$(f/g)(x) = f(x)/g(x)$

Definition (Domain of Operations): For the sum, difference, and product, the domain is the intersection of the domains of f and g. For the quotient, the domain excludes points where $g(x) = 0$.

Definition (Composition):

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ consists of all x in the domain of g for which $g(x)$ is in the domain of f.

Example: $f(x) = x^2 + 3$, $g(x) = \sqrt{x}$

$$(f \circ g)(x) = (\sqrt{x})^2 + 3 = x + 3$$

Domain: $[0, +\infty)$

Important: $(f \circ g) \neq (g \circ f)$ generally.

Example: Show $f \circ g \neq g \circ f$ for $f(x) = x^2$, $g(x) = x + 1$.

$$(f \circ g)(x) = f(g(x)) = (x+1)^2$$
$$(g \circ f)(x) = g(f(x)) = x^2 + 1$$

Since $(x+1)^2 \neq x^2 + 1$, the composition is not commutative.

Definition (Notation for Powers):

$$f^2(x) = f(x) \cdot f(x), \quad f^n(x) = \underbrace{f(x) \cdot f(x) \cdots f(x)}_{n \text{ times}}$$

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Definition (Decomposition): Breaking a function into simpler ones.

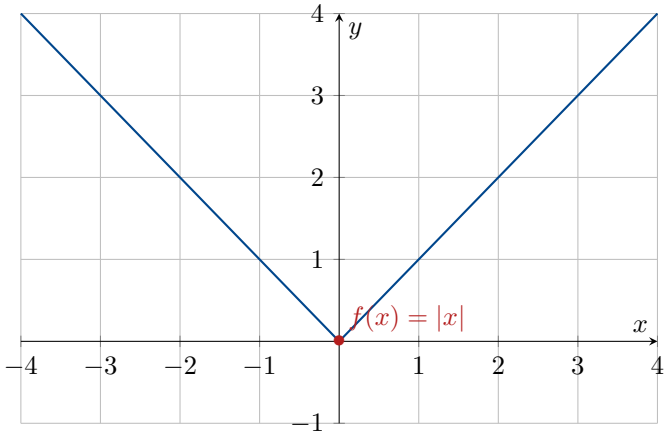
$$h(x) = (x+1)^2 \Rightarrow f(x) = x+1, \quad g(x) = x^2, \quad h(x) = g(f(x))$$

2 Lesson 08: Graphing Functions

Definition (Graph of a Function): The graph of a function f in the xy-plane is the graph of the equation $y = f(x)$.

Example: Sketch $f(x) = |x|$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



Definition (Translations):

Transformation	Geometric Effect
$y = f(x) + c$	Translation **UP** by c units
$y = f(x) - c$	Translation **DOWN** by c units
$y = f(x + c)$	Translation **LEFT** by c units
$y = f(x - c)$	Translation **RIGHT** by c units

Definition (Reflections):

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Transformation	Geometric Effect
$y = f(-x)$	Reflection about y-axis
$y = -f(x)$	Reflection about x-axis

Definition (Vertical Scaling): If $f(x)$ is multiplied by a positive constant c:

- Compressed vertically if $0 < c < 1$
- Stretched vertically if $c > 1$

Vertical Line Test: A graph is the graph of a function if and only if NO vertical line intersects the graph more than once.

3 Lesson 09: Limits (Intuitive Introduction)

Note (Two Branches of Calculus):

- Calculus from the tangent problem → Differential Calculus
- Calculus from the area problem → Integral Calculus

Both depend on the fundamental notion of Limit.

Definition (Limit): Limits study the behavior of the y-values of a function in response to the x-values as they approach some number or go to infinity.

Definition (Notation):

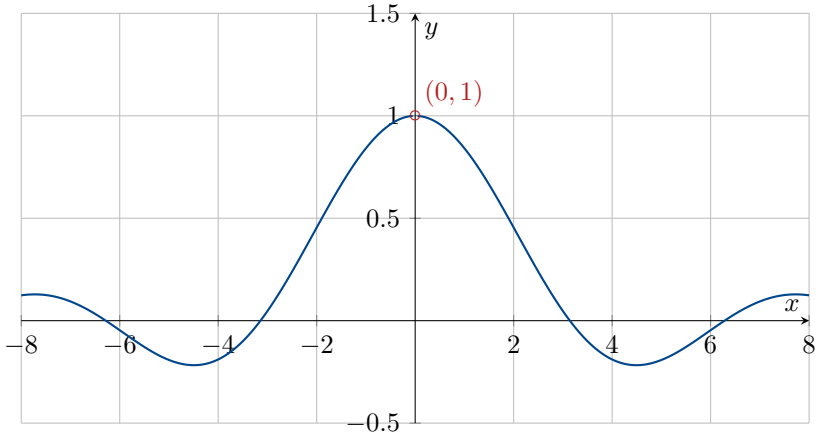
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Notation	Meaning
$\lim_{x\to a^+} f(x) = L$	Limit as $x \to a$ from the right
$\lim_{x\to a^-} f(x) = L$	Limit as $x \to a$ from the left
$\lim_{x\to a} f(x) = L$	Limit as $x \to a$ (both sides)
$\lim_{x\to +\infty} f(x) = L$	Limit as $x \to +\infty$
$\lim_{x\to -\infty} f(x) = L$	Limit as $x \to -\infty$

Example: $\lim_{x\to 0} \dfrac{\sin(x)}{x}$

$$\lim_{x\to 0^+} \frac{\sin(x)}{x} = 1$$
$$\lim_{x\to 0^-} \frac{\sin(x)}{x} = 1$$

When both match: $\lim_{x\to 0} \dfrac{\sin(x)}{x} = 1$



Definition (Existence of Limit): A limit does not exist (DNE) due to:

1. Oscillations
2. Unbounded increase or decrease

Example (Unbounded increase):

$$\lim_{x\to x_0^+} f(x) = \lim_{x\to x_0^-} f(x) = \lim_{x\to x_0} f(x) = +\infty$$

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The limit FAILS to EXIST because of unboundedness. The $+\infty$ is NOT a number!

4 Lesson 10: Limits (Computational Techniques)

Definition (Basic Limits):

Limit	Value
$\lim_{x\to a} k$	k
$\lim_{x\to a} x$	a
$\lim_{x\to+\infty} x$	$+\infty$
$\lim_{x\to-\infty} x$	$-\infty$

Theorem 2.5.1: If $L_1 = \lim f(x)$ and $L_2 = \lim g(x)$ both exist, then:

$$\begin{aligned}\lim[f(x)+g(x)] &= L_1 + L_2\\ \lim[f(x)-g(x)] &= L_1 - L_2\\ \lim[f(x)\cdot g(x)] &= L_1 \cdot L_2\\ \lim\frac{f(x)}{g(x)} &= \frac{L_1}{L_2} \quad (L_2 \neq 0)\end{aligned}$$

Key Results:

$$\lim_{x\to a}(x^n) = a^n, \quad \lim[k\cdot g(x)] = k\cdot\lim g(x)$$

A constant factor can be moved through a limit sign.

Example: $\lim_{x\to 5}(x^2 - 4x + 3)$

$$= (5)^2 - 4(5) + 3 = 25 - 20 + 3 = 8$$

Definition (Limits Involving $1/x$):

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Limit	Value
$\lim_{x\to 0^+} \frac{1}{x}$	$+\infty$
$\lim_{x\to 0^-} \frac{1}{x}$	$-\infty$
$\lim_{x\to \pm\infty} \frac{1}{x}$	0

Example: $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$

$$= \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2}(x+2) = 4$$

Definition (Quick Rule for $\pm\infty$):

$$\lim_{x\to\pm\infty} \frac{c_0 + c_1 x + \cdots + c_n x^n}{d_0 + d_1 x + \cdots + d_m x^m} = \lim_{x\to\pm\infty} \frac{c_n x^n}{d_m x^m}$$

Not true if $x \to a$ (finite number).

Example: $\lim_{x\to -\infty} \dfrac{4x^2 - x}{2x^3 - 5}$

$$= \lim_{x\to -\infty} \frac{4x^2}{2x^3} = \lim_{x\to -\infty} \frac{2}{x} = 0$$

5 Lesson 11: Limits (Rigorous Approach)

Definition (Formal Definition of Limit): For any number $\epsilon > 0$, if we can find an open interval (x_0, x_1) containing a such that

$$L - \epsilon < f(x) < L + \epsilon$$

for each x in (x_0, x_1) except possibly $x = a$, then we say

$$\lim_{x\to a} f(x) = L$$

Definition (Equivalent Forms):

$$|f(x) - L| < \epsilon \quad \text{and} \quad 0 < |x - a| < \delta$$

Example: Find δ such that $\lim_{x\to 2}(3x - 5) = 1$.

We need to find $\delta > 0$ such that

$$|(3x-5) - 1| < \epsilon \quad \text{whenever} \quad 0 < |x - 2| < \delta$$

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Step 1: Simplify the expression.

$$\begin{aligned} |(3x-5)-1| &< \epsilon \\ |3x-6| &< \epsilon \\ 3|x-2| &< \epsilon \\ |x-2| &< \frac{\epsilon}{3} \end{aligned}$$

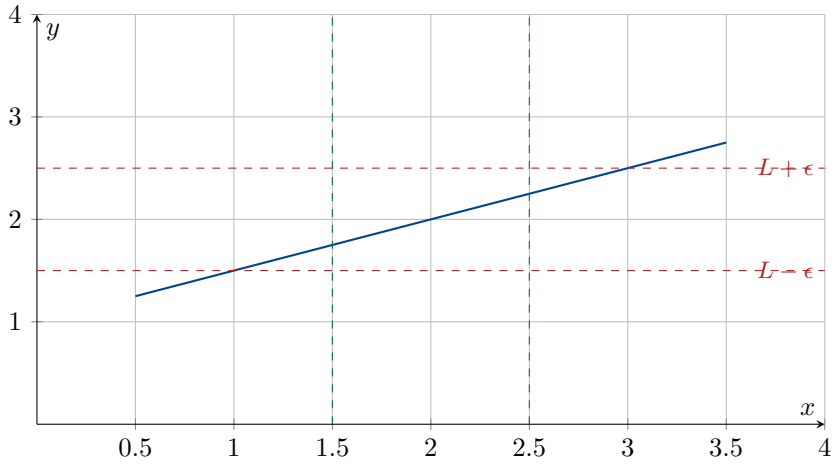
Step 2: Choose δ.

$$\boxed{\delta = \frac{\epsilon}{3}}$$

Step 3: Verify. If $0 < |x-2| < \delta = \epsilon/3$, then

$$|(3x-5)-1| = 3|x-2| < 3 \cdot \frac{\epsilon}{3} = \epsilon$$

Hence $\lim_{x\to 2}(3x-5) = 1$. ■



Example: Show $\lim_{x\to 0} f(x)$ does not exist for

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Suppose the limit exists and equals L. Take $\epsilon = 1$. There is $\delta > 0$ such that:

$$|f(x) - L| < 1 \quad \text{if } 0 < |x| < \delta$$

But $x = \delta/2$ and $x = -\delta/2$ both satisfy this, so:

$$|1 - L| < 1 \quad \text{and} \quad |-1 - L| < 1$$

$$0 < L < 2 \quad \text{and} \quad -2 < L < 0$$

Contradiction! So the limit does not exist. ■

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6 Lesson 12: Continuity

Definition 2.7.1 (Continuity): A function f is continuous at $x = c$ if:

(a) $f(c)$ is defined

(b) $\lim_{x \to c} f(x)$ exists

(c) $\lim_{x \to c} f(x) = f(c)$

If any condition fails, f is **discontinuous** at c.

Example: $f(x) = \dfrac{x^2 - 4}{x - 2}$
Discontinuous at $x = 2$ because $f(2)$ is undefined.

Example: $g(x) = \begin{cases} \dfrac{x^2 - 4}{x - 2} & x \neq 2 \\ 3 & x = 2 \end{cases}$

$$\lim_{x \to 2} g(x) = \lim_{x \to 2} (x + 2) = 4 \neq g(2) = 3$$

So g is discontinuous at $x = 2$.

Theorem 2.7.2: Polynomials are continuous functions.

Theorem 2.7.3 (Properties): If f and g are continuous at c, then:

Property	Condition
$f + g$ is continuous at c	—
$f - g$ is continuous at c	—
$f \cdot g$ is continuous at c	—
f/g is continuous at c	$g(c) \neq 0$

Theorem 2.7.5 (Composition): If $\lim g(x) = L$ and f is continuous at L:

$$\lim f(g(x)) = f(L)$$

Definition 2.7.7 (One-Sided Continuity):

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Continuous from the left at c:

1. $f(c)$ is defined
2. $\lim_{x \to c^-} f(x)$ exists
3. $\lim_{x \to c^-} f(x) = f(c)$

Continuous from the right at c:

1. $f(c)$ is defined
2. $\lim_{x \to c^+} f(x)$ exists
3. $\lim_{x \to c^+} f(x) = f(c)$

Definition 2.7.8 (Continuity on $[a, b]$): f is continuous on $[a, b]$ if:

1. f is continuous on (a, b)
2. f is continuous from the right at a
3. f is continuous from the left at b

Theorem 2.7.9 (Intermediate Value Theorem): If f is continuous on $[a, b]$ and C is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number x in $[a, b]$ such that $f(x) = C$.

Theorem 2.7.10: If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there is at least one solution of $f(x) = 0$ in (a, b).

End of Part 1B (Lessons 07–12)

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